

Understanding Maven for CI/CD

Success 🚀

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What is Maven? 🤔

Imagine you're building a LEGO castle. You need:

- A **plan** 📖 (like instructions),
- The same **bricks** every time 🧱,
- And you want to build the same perfect castle **again and again** 🔁.

Maven is like that for Java (and some other) projects.

- 🧰 It's a **build tool**: It helps put together all the code, files, and settings needed to make your app work.

- 📦 It **downloads all the libraries** (called dependencies) you need from the internet — so you don't go hunting for them.
- 🏗️ It **builds** your code into a working product (like `.jar` or `.war` files).
- 🛠️ It also runs **tests**, checks for bugs, and even **packages** your app.

Maven uses a file called `pom.xml` where all the rules are written — like what version of Java to use, what libraries are needed, and what to do when we build or test the app.

Maven in DevOps Infra

In DevOps, we care about:

- CI Pipelines 

- Artifact Repositories 

- Version Control Systems (VCS) 

Let's see how Maven fits in all of them 

Maven in CI Pipelines

In **Continuous Integration (CI)**, code is tested every time someone changes it. Here's where Maven helps:

Example — Jenkinsfile:

```
stage('Build') {  
  steps {  
    sh 'mvn clean install'  
  }  
}
```

💡 This line tells Jenkins to use Maven to clean old builds and build a new one. Maven takes care of downloading libraries and compiling everything properly.

Artifact Repositories

After building, we need to **store the result** somewhere — like a **.jar** file.

That's where **artifact repositories** like **Nexus** or **JFrog Artifactory** come in.

Maven uploads artifacts (like **.jar**) here, so others or future builds can download them anytime. 

Example — pom.xml snippet

```
<distributionManagement>
  <repository>
    <id>my-nexus</id>
    <url>https://nexus.myorg.com/repository/releases</url>
  </repository>
</distributionManagement>
```

 This tells Maven: "After building, send it to Nexus!"

Version Control Systems (VCS)

When code is in **Git** (or GitHub), every change should trigger:

- A **build**,
- A **test**, and
- A **new artifact**, if all is good 

Maven makes that easy: It does the heavy lifting so DevOps tools like GitHub Actions or Jenkins don't have to worry about "how to build".

Why is Maven Everywhere?

-  **Standardized**: Same structure, same steps — no confusion.

-  **Automates everything:** From build to test to package.
-  **Works with other DevOps tools:** Jenkins, Nexus, Git — they all ❤️ Maven.
-  **Scalable:** Big companies use it because it works great for big projects too.

 **Fun Fact:** Most Java projects in big companies use Maven daily in their pipelines!

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